DESN2000 SOFTWARE ENGINEERING

Acceptance Criteria

Prototyping

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Feedbacks from students

- None, which makes me wonder ☺

What are we doing?

1. Requirement Validation
    - Acceptance Criteria

2. Some Case Studies

3. Prototyping
    - What is prototyping
    - Low/High Fidelity and everything in between

4. FIGMA
    - Welcome to Figma

Requirements Validation

User Story

Acceptance criteria

- When is a user story complete?
  - Acceptance criteria
  - Dependencies
  - UX Artefacts

What?

Acceptance criteria

- Acceptance criteria are usually a set of statements
- Each statement should have a **clear pass/fail** result,
- Each statement should be **measurable**
- Criteria should specify both **functional** and **non-functional** requirements.

- The goal of acceptance criteria is to **explain** the conditions that a specific user story must satisfy

What?

Acceptance criteria

- Document prior to development
- Don’t make it too narrow
- Keep it achievable
- Keep it measurable and not too broad
- Avoid technical details
- Collaborate and reach consensus
- Make it testable
- Active-first person voice
- Avoid negative sentences
- Simple and concise

Why?

Acceptance criteria

- Can help avoid unexpected results at the end of development
- Ensures all stakeholders and users are getting what they want

- They have to **be defined before any development** takes place!

Why?

Acceptance criteria

Defines boundaries of user stories

Helps to see negative scenarios

Establish a common language for the whole team and the stakeholders

Streamline the testing process

Agile Testing:
Behaviour Driven Development

Gherkin

- Scenario:

- Given (context/an initial condition)
- And (more context if needed)
- When (event/something happens)
- Then (outcome/this should be the result) And (another outcome)

https://cucumber.io/docs/guides/overview/

Agile Testing:
Behaviour Driven Development

Example

Feature: Searching on Google

As a web surfer, I want to search Google, so that I can learn new things.

- Scenario: Simple Google search
- GIVEN a web browser is on the Google homepage
- WHEN the search phrase “Agile” is entered
- THEN results for “Agile” are shown

Agile Testing: Behaviour Driven Development

Example

Feature: Searching on Google

User Story: As a web surfer, I want to search Google, so that I can learn new things.

- Scenario: Simple Google search
- GIVEN a web browser is on the Google homepage WHEN the search phrase “Agile” is entered
- THEN results for “Agile” are shown
- AND the related results include “Agile testing”
- BUT the related results do not include “Agile running”

Agile Testing: Behaviour Driven Development

Example

User story: As a user, I want to be able to recover the password to my account, so that I will be able to access my account in case I forgot the password.

- Scenario: Forgot password

- Given: The user has navigated to the login page
- When: The user selected forgot password option
- And: Entered a valid email to receive a link for password recovery
- Then: The system sends the link to the entered email

Agile Testing: Behaviour Driven Development

Example

User story: As a user, I want to be able to recover the password to my account, so that I will be able to access my account in case I forgot the password.

- Scenario: Forgot password

- Given: The user received the link via the email
- When: The user navigated through the link received in the email
- Then: The system enables the user to set a new password

Agile Testing: Behaviour Driven Development

Example

User story: As a user, I want to be able to recover the password to my account, so that I will be able to access my account in case I forgot the password.

- Scenario: Forgot password

- Given: the Form is submitted
- When: the value entered in the Number text box is not numerical
- Then: an error message “Please enter a numerical value” appear

Agile Testing: Behaviour Driven Development

Example

User story: As a user, I want to be able to recover the password to my account, so that I will be able to access my account in case I forgot the password.

- Scenario: Forgot password

- Given: the value entered in the Number text box is not numerical
- When: the Form is submitted
- Then: an error message “Please enter a numerical value” appear.

Agile Testing: Behaviour Driven Development

Example

User story: As a user, I want to be able to recover the password to my account, so that I will be able to access my account in case I forgot the password.

- **Scenario:** Forgot password

- **Given:** the Form is submitted
- **When:** the value entered in the Number text box is not numerical
- **Then:** an error message "Please enter a numerical value" appear

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Agile Testing:
Behaviour Driven Development

Scenario	This is the title of the condition to be acted upon.
Given (an initial condition)	The user places an item into their shopping cart.
When (something happens)	This is where the process in which the user's initial order is verified or whether it fulfills the system requirements to process the task. If it does, then the system can proceed to work on the order. However, if the user order does not match to the system requirements, the system will deny the task.
Then (this is the result)	Once the system is done verifying the user order, the order is then processed to produce the results which would be: the final result, input to the next task or a lead-on for the user to the next task.

Great, What Happens If I Can’t Fit Into **Given/When/Then**?

Rule oriented acceptance criteria!

A set of rules that describe the behaviour of a system

Example: As a user, I want to use a search field to type a first name, surname, or job description, so that I could find matching staff members.

What could be some rules that we work with for this user story?

Consider Non-Functional Requirements As User Stories Also

Can you think of an example user story that would encompass a non-functional requirement?

Given I am on the contacts page
And I have refreshed the page
When refreshing takes more that the max acceptable time
Then the problem is logged

https://cucumber.io/blog/hiptest/how-to-address-non-functional-requirements/

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Let's try Gherkin to write a scenario for "As a student, I want to be able to raise my hand virtually if I have a question, so that the lecturer/tutor can answer my question "

ⓘ Start presenting to display the poll results on this slide.

When Testing Fails

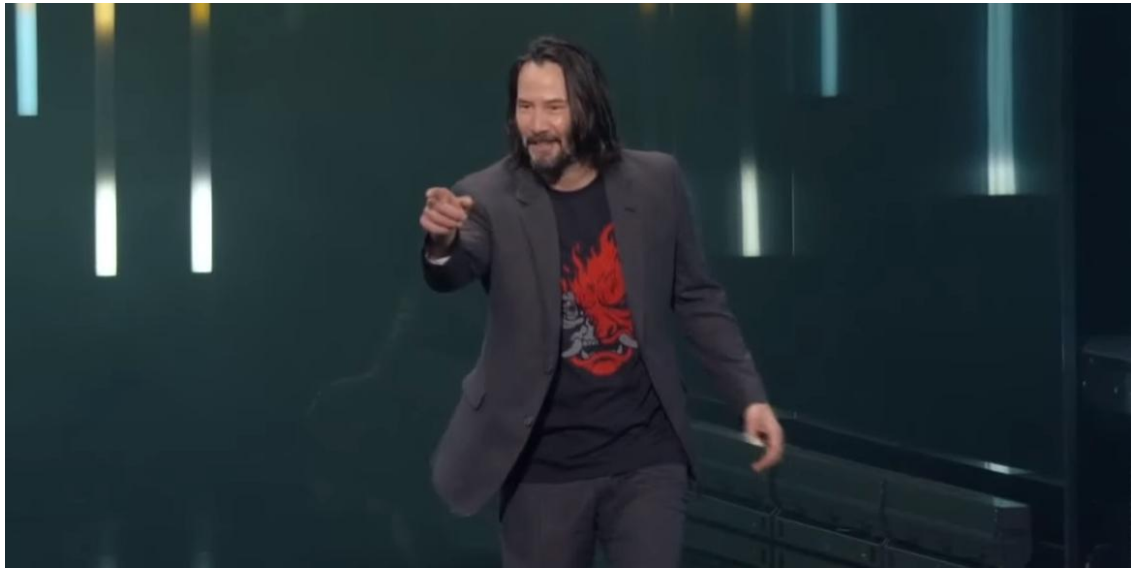
Case Studies

Nest Thermostat

Example

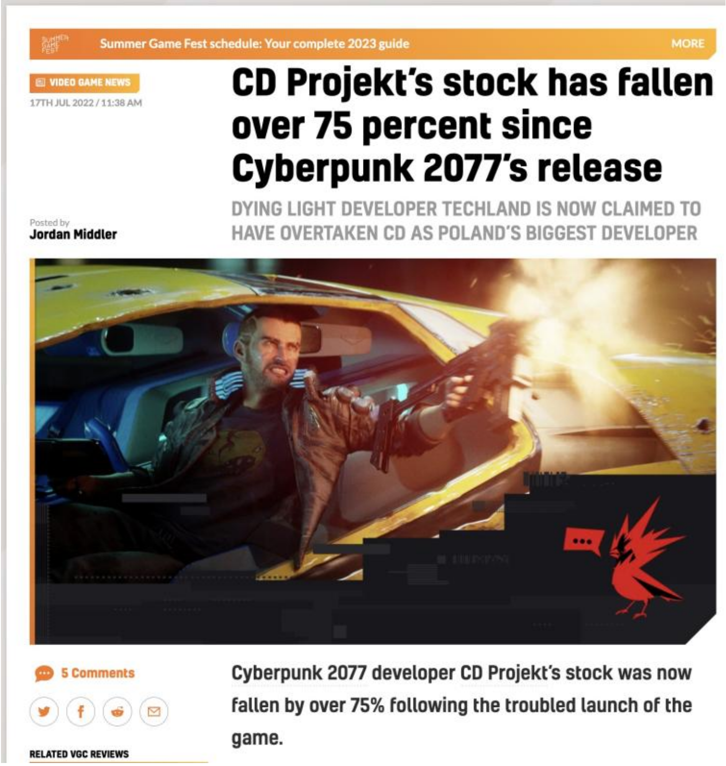
https://www.bbc.com/news/technology-35311447

Cyberpunk 2077



Cyberpunk 2077

Cyberpunk 2077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VIDEO GAME NEWS

17TH JUL 2022 / 11:38 AM

CD Projekt's stock has fallen over 75 percent since Cyberpunk 2077's release

DYING LIGHT DEVELOPER TECHLAND IS NOW CLAIMED TO HAVE OVERTAKEN CD AS POLAND'S BIGGEST DEVELOPER

Posted by
Jordan Middler

5 Comments

Cyberpunk 2077 developer CD Projekt's stock was now fallen by over 75% following the troubled launch of the game.

RELATED VGC REVIEWS

https://www.videogameschronicle.com/news/cd-projekts-stock-has-fallen-over-75-percent-since-cyberpunk-2077s-release/

Share with us some examples… use the Forum in Teams if you like.

Prototyping Software Products

What is Prototyping?

A simulation of the real system, allows us to do early testing to see whether our prototype can meet user goals

What is Prototyping?

- Prototypes will be based on the previous user centered design activities
  - Product objectives
  - User research
  - Scenarios/Journey Maps
  - Problem Statements
  - Your user stories and acceptance criteria

What is Prototyping?

- Sketches
- Storyboards
- Low Fidelity
- Wizard of Oz
  - a prototype that works by having someone behind-the-scenes who is pulling the levers and flipping the switches
- 3D object (physical devices, Palm)
- High Fidelity

What is Prototyping?

- Evolutionary Prototyping
  - Each prototype is a real piece of the final product
- Throw away prototype
  - Prototypes don’t go into production systems

What is Prototyping?

- Horizontal
  - broad, lots of functions
- Vertical
  - analyse functionality quite deeply, not many functions

What is Prototyping?

- Prototyping is iterative
- Communicate aspects of the design

What is Prototyping?

Fidelity

- Different stages of design require different levels of complexity, difficulty, and fidelity.

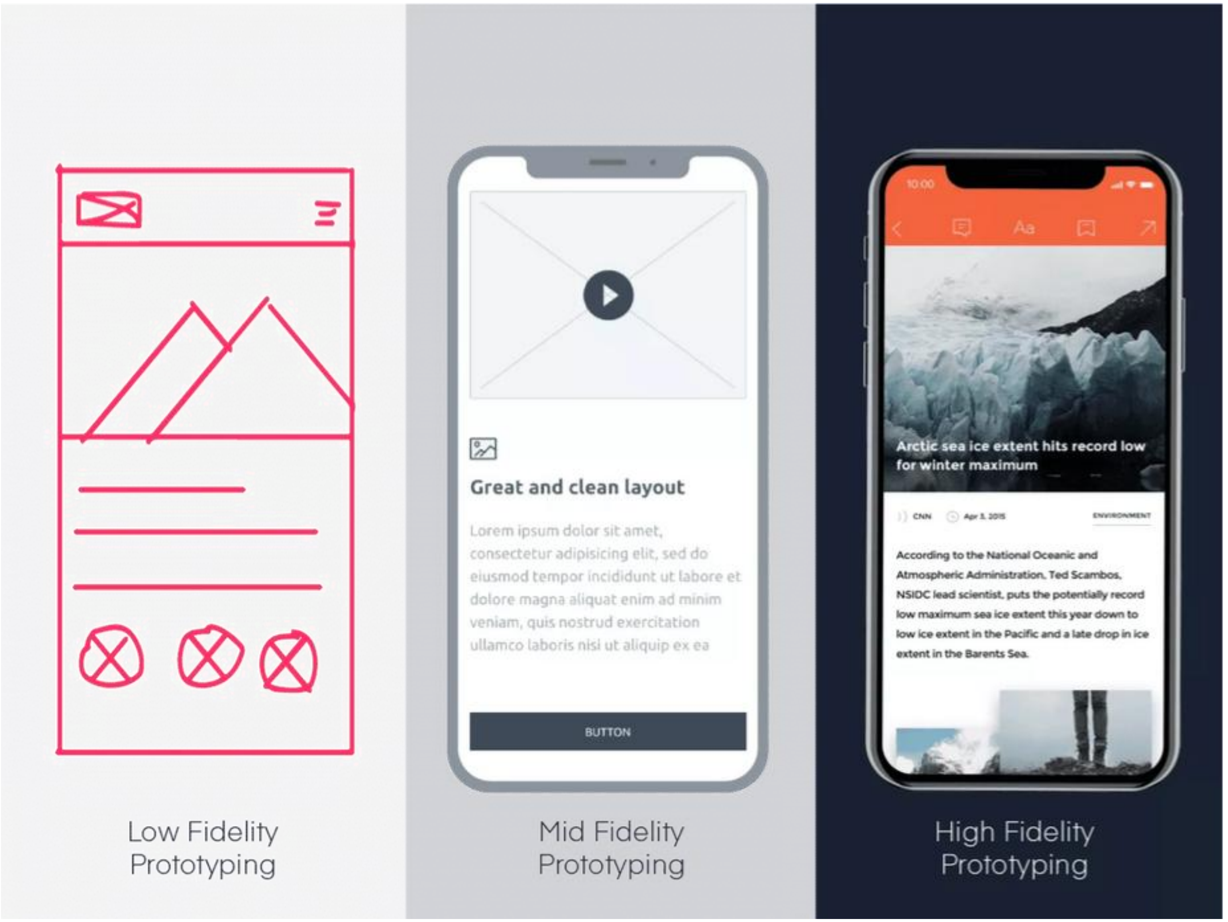


image Credit: https://proportion.global/prototyping/

What is Prototyping?

Fidelity

- Fidelity refers to how closely a prototype resembles a final solution:
    - Visual (sketched -> styled)
    - Functional (static -> interactive)
    - Content (lorem ipsum -> real content)

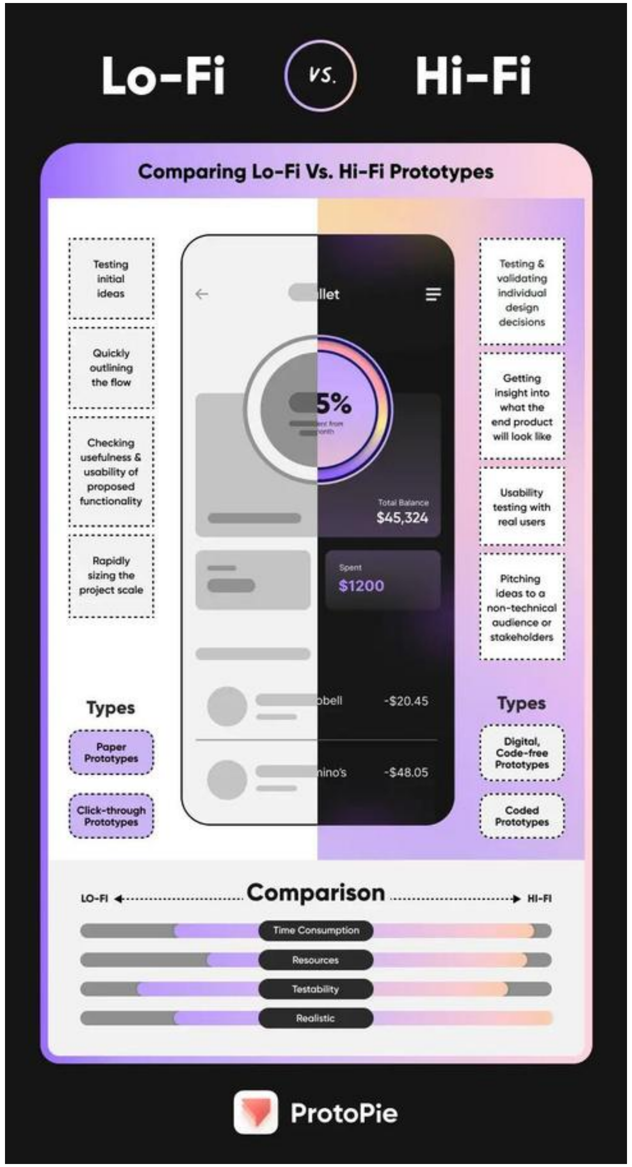


Image Credit: https://www.protopie.io/blog/l ow-fidelity-vs-high-fidelity- prototyping

What is Prototyping?

Fidelity

- Choosing a fidelity

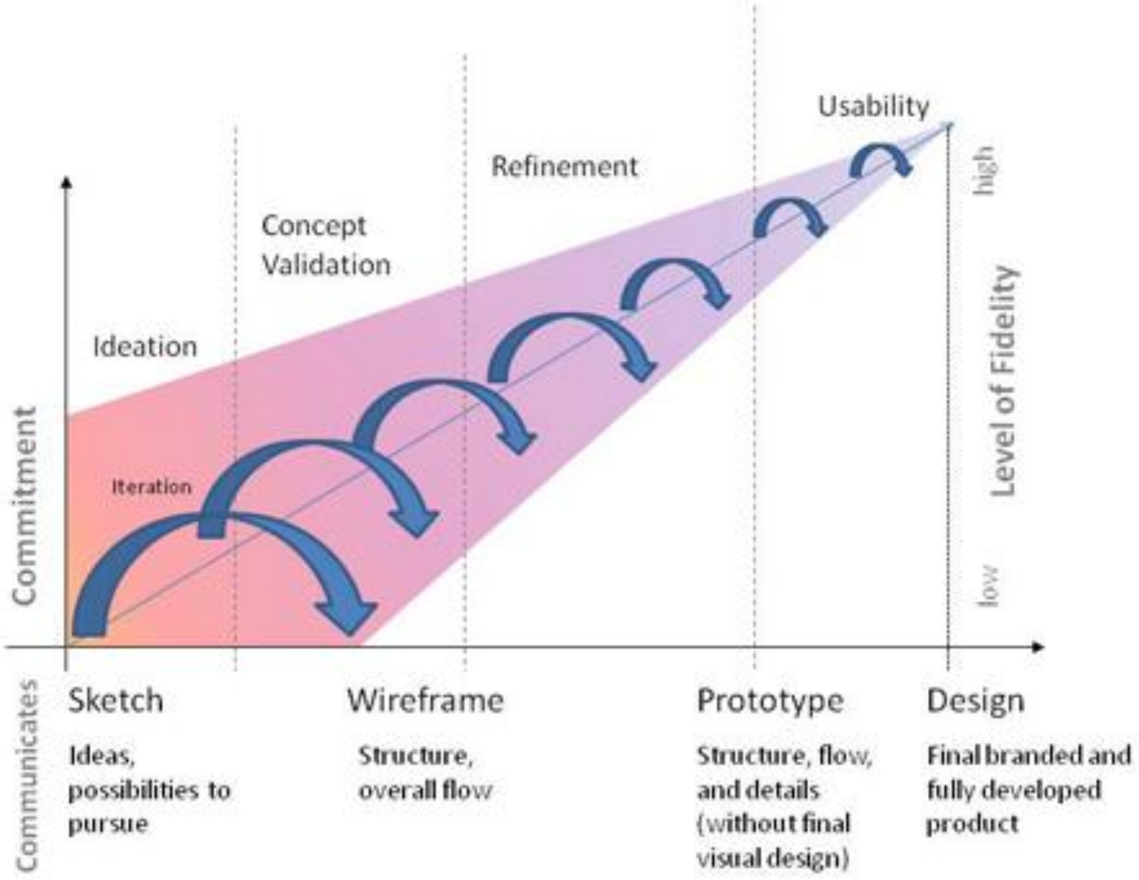
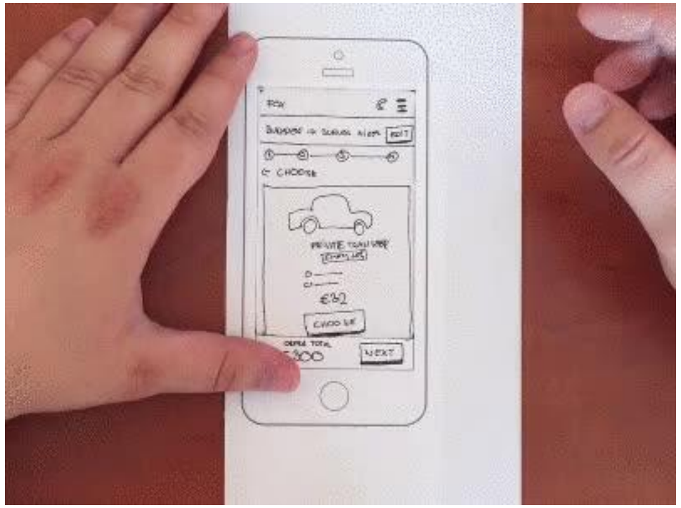


Image credit: https://www.uxmatters.com/mt/archives/2010/05/sketches-and-wireframes-and-prototypes-oh-my-creating-your-own-magical-wizard- experience.php

Low Fidelity Prototyping

First – paper!



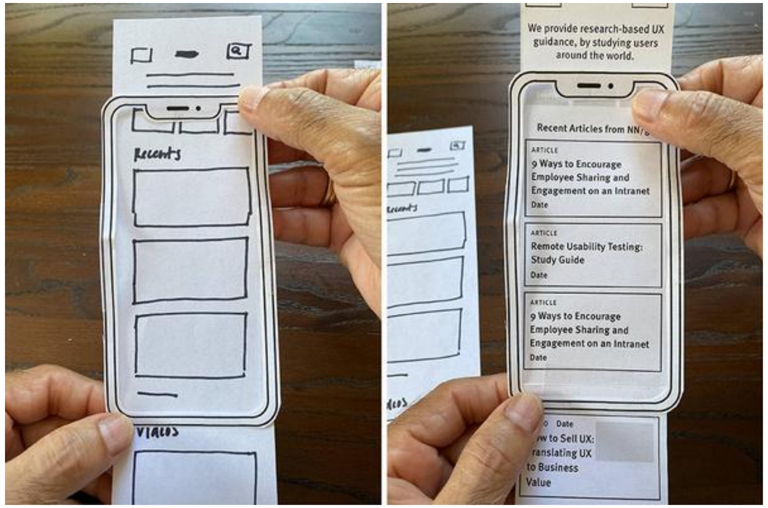


Image credit: https://dribbble.com/shots/14549067- Low-Fidelity-Prototype

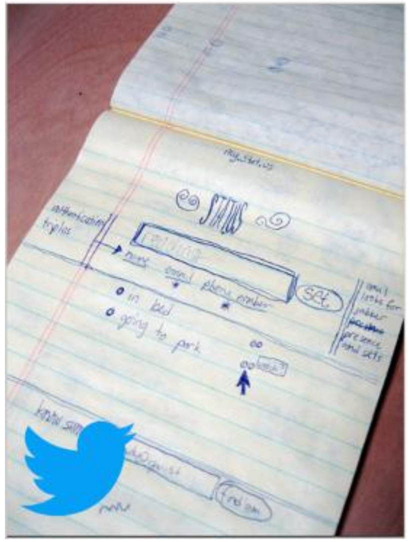
Image credit: https://www.sitepoint.com/how-to- make-paper-prototypes/

Image credit: https://www.nngroup.com/articles/paper-prototyping-cutout-kit/

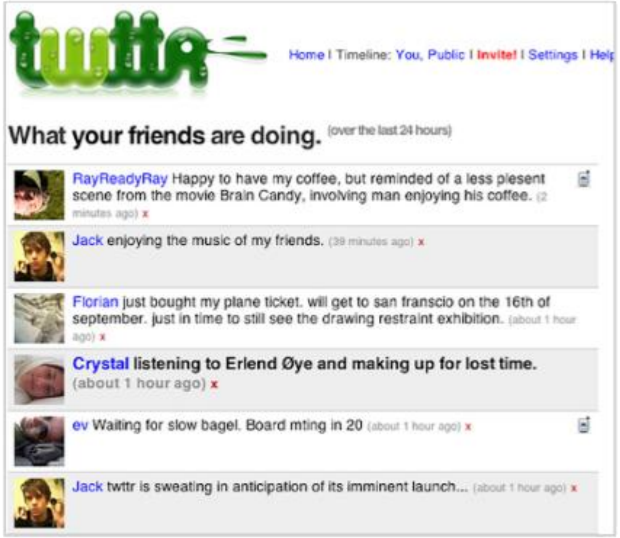
Twitter

How Twitter started versus how it is going now

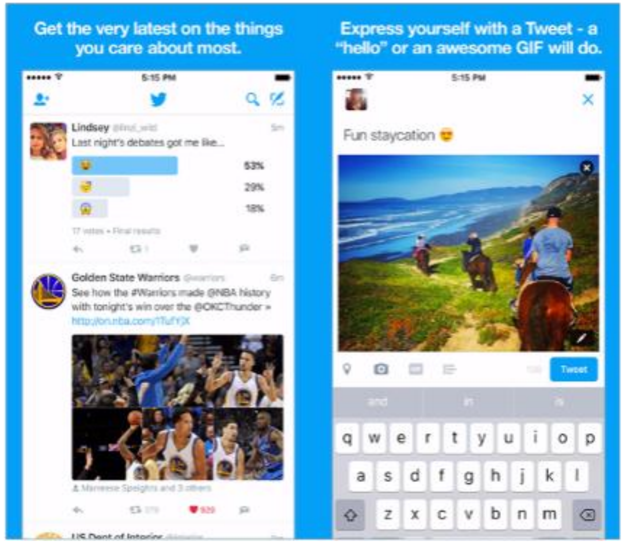
2000



2006



2017



Low Fidelity Prototyping

Paper

- Advantages:
  - Fast way of prototyping
  - Highly collaborative
  - Testing
  - Promotes experimentation
  - Lowers the bar
  - Communication
- Can establish terminology (and logical) flaws very early in design
- Designing and thinking through main user flows
- Determining key interactions

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What are the disadvantages of using Paper Prototyping?

ⓘ Start presenting to display the poll results on this slide.

Low Fidelity Prototyping

Paper

- Disadvantages:
    - Hard to interpret design
    - Testing can only happen in person
    - Less useful near the end of a production cycle Animations
    - Need to transfer to digital format at some stage

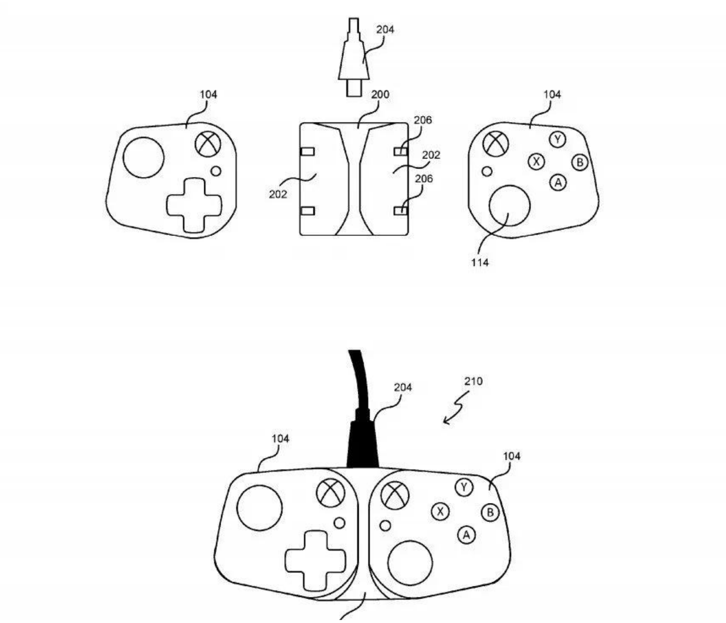
Why Bother?

Paper Prototypes

- Do the users actually do things the way that the design currently works?
- Does the team really understand what the user wants/needs?

Low Fidelity In Real Life

Xbox Controller



Moving Up in the Fidelity Stakes

High Fidelity Prototypes

- Develop proof of concept high fidelity prototypes that help explore the interactivity of the system
- Could be done as part of the software development process
  - Should be considered to be an ‘interim step’ toward a final design – not the final design!
- Could be done with tools that can animate the future system
  - Closely related to the concept of storyboarding

Moving Up in the Fidelity Stakes

High Fidelity Prototypes

- Advantages:
  - Specific, realistic
  - More valuable to developers than paper
  - Realism can help project success
  - Can incorporate technologies that Paper cannot (multi-touch, animation, etc)

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What are the disadvantages of using High Fidelity prototyping?

ⓘ Start presenting to display the poll results on this slide.

Moving Up in the Fidelity Stakes

High Fidelity Prototypes

- Disadvantages:
  - Slower to iterate (depending on toolset and competence)
  - Information Overload if used too early
  - Costly (skillsets, time - cost rises with precision)
  - Can be unreliable as interactivity is added

Moving Up in the Fidelity Stakes

High Fidelity Prototypes

Video Credit: https://dribbble.com/shots/16213456--oncept-for-a-mass-market-%20vehicle-HMI

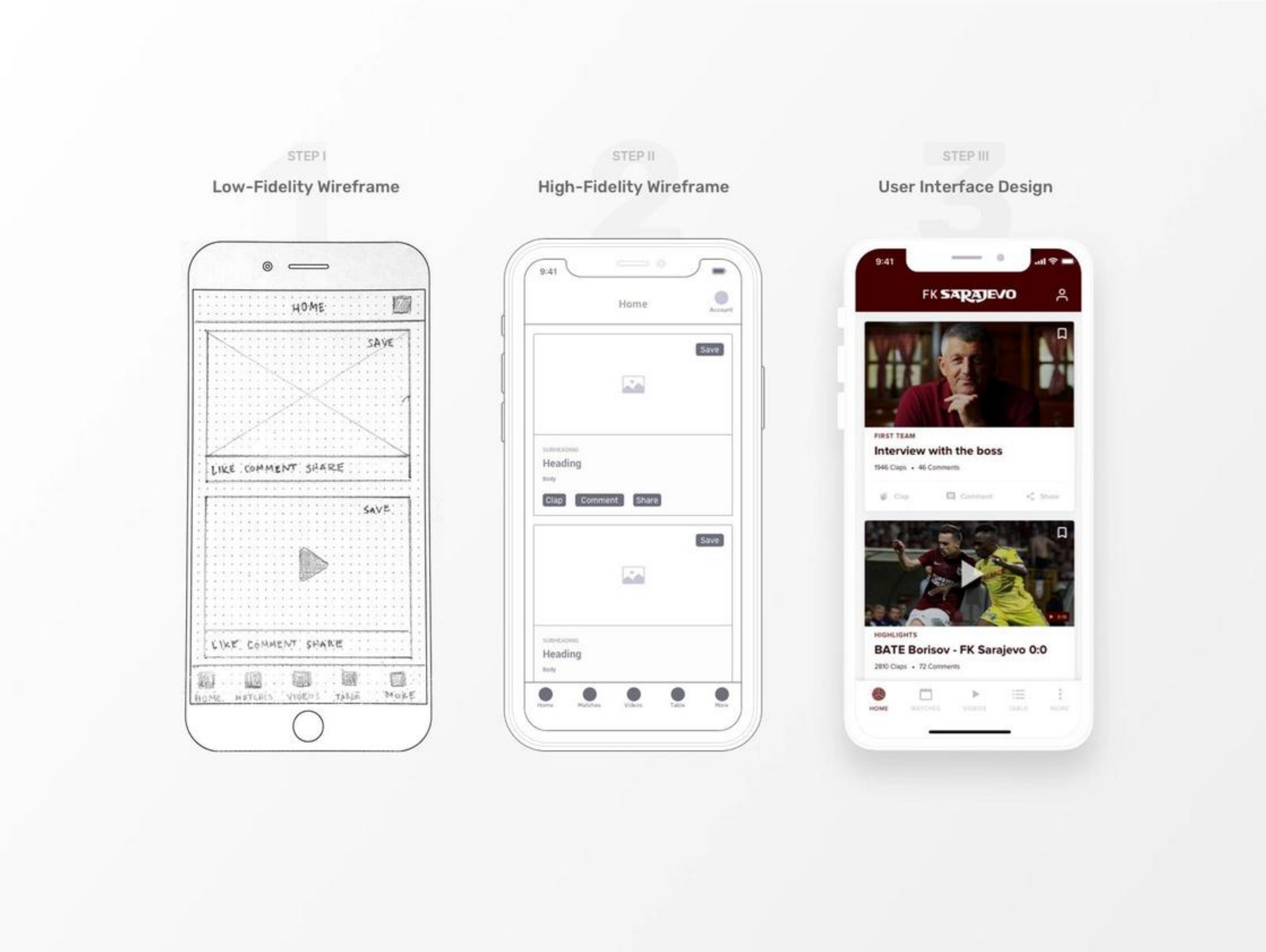
In Summary

Prototyping - Sketches to low fidelity to high fidelity

1. Define the vision (sketches)
   a) What problem does it solve?
   b) Who is the target demographic?
   c) What are other alternatives?
2. Focus on key features (low fidelity prototyping)
   a) This is not the final product!
3. Test and Refine (move up the fidelity ladder)
   a) You may scrap it all at this point and start again! This is still an excellent outcome.

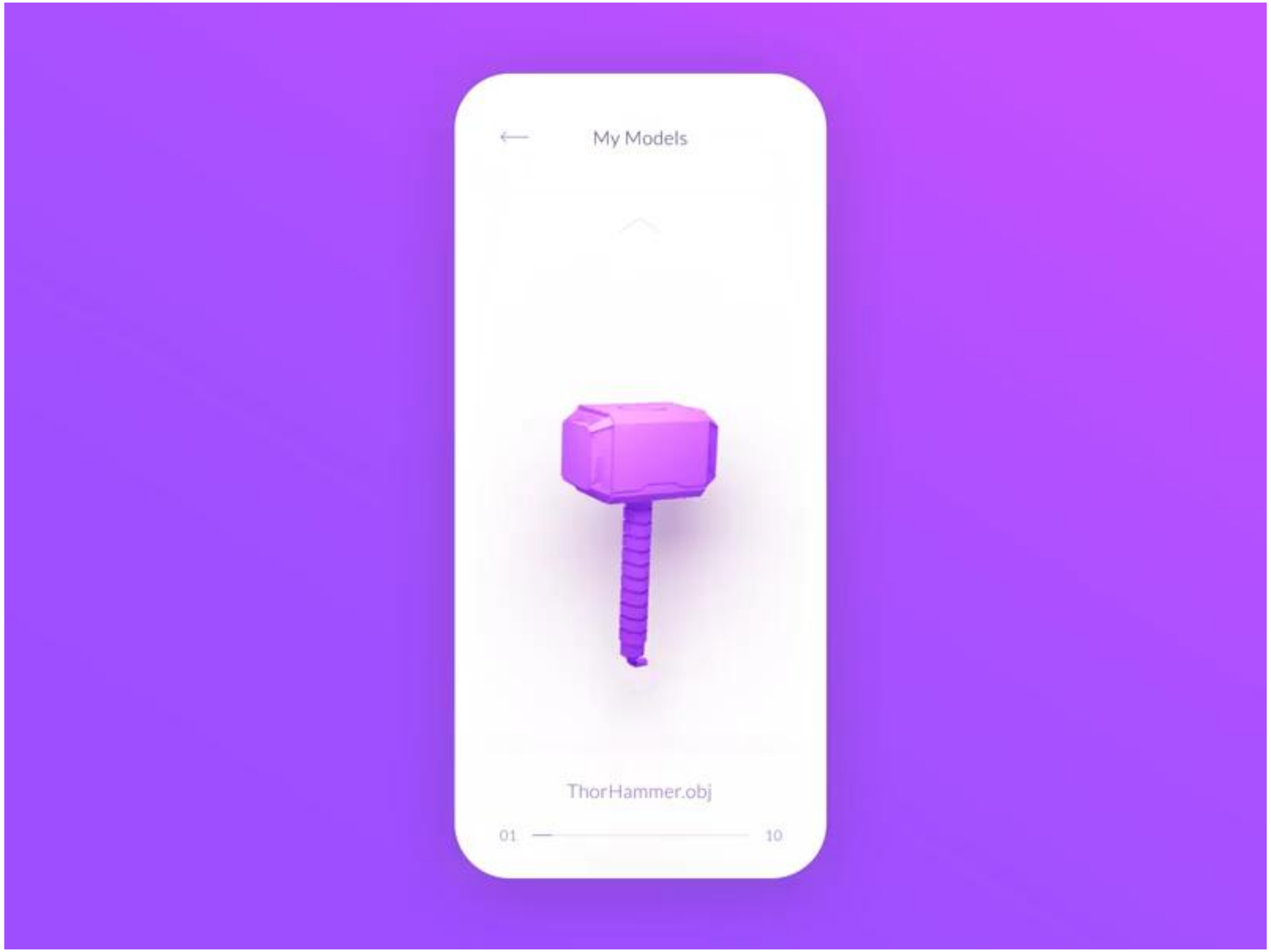
Case Studies

Prototyping –
Sketches to low fidelity to high fidelity



Case Study

High Fidelity



Case Studies

Low Fidelity

Case Studies

Somewhere in between

Microsoft Surface

Galaxy Fold

Tools for Prototyping

Prototyping Tools

Many!

- Balsamiq (low-fidelity)
- Adobe XD
- Figma
- Sketch
- JustinMind
- InVision
- Axure
- The New Kid in the Block (your Favourite AI Tool)

Prototyping Tools

Figma

- Short demo video for Figma Jira Integration (https://youtu.be/91nsuY0a6uA?feature=shared)
- Short Video for using Figma for Collaboration and handoff (https://www.youtube.com/watch?v=xCJsRuH7v9w)
- Wireframing in Figma (https://www.youtube.com/watch?v=qWIdforZ9x0)

Requirements Elicitation in the Age of GenAI

What changes when Gen AI enters requirements elicitation?

The activity becomes less about generating ideas and more about validating, grounding, and constraining AI output.

User research → Themes & insights → Requirements → AI prompt → Generated code → Validation

In Requirements Elicitation AI helps with speed

Draft interview questions, summarise transcript notes, cluster themes, fix error in text.

Humans own validity

Access to users, observation of real behaviour, domain judgement, ethics, evidence and final acceptance remain human responsibilities.

The new bottleneck

The quality of the prompt is limited by the quality of the elicited requirement. A weak requirement simply automates the wrong assumption faster.

GenAI shifts the “hard part” upstream. It reduces typing effort, but increases the value of clear problem framing.

Reference: https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2025.1519437/full

Vibe coding: where requirements become the control surface

- What?
  - ➢ Describe the desired system to an AI tool
  - ➢ Let the tool generate code or UI quickly
  - ➢ Iterate by asking for changes instead of manually rewriting everything]

- Why is it Attractive?
  - ➢ Very fast prototypes
  - ➢ Accessible to non-programmers
  - ➢ Useful for exploring interface ideas and workflows
  - ➢ Good for throwaway experiments

- How is it Risky?
  - ➢ Hidden assumptions become hidden code
  - ➢ Non-functional requirements are often omitted
  - ➢ Users may accept code they do not understand
  - ➢ Testing and security are easy to skip
  - ➢ Faster Pace development makes missing requirements harder to notice

Do high-quality requirements lead to better AI-generated code?

Yes — because requirements become the prompt, the constraints and the test oracle.

Low quality

“Build a login screen.”

Likely AI behaviour:
• assumes user types
• skips edge cases
• may omit security
• hard to test

Better

“Students must be able to sign in using UNSW credentials and recover access if they forget their password.”

Better, but still missing constraints.

AI-ready

User story + acceptance criteria + constraints + error states + NFRs + test examples.

The AI has a much smaller space for unsafe assumptions.

AI code quality improves when the requirement contains: role, goal, context, data, constraints, acceptance criteria, and non-functional requirements.

Reference: https://github.blog/news-insights/research/research-quantifying-github-copilots-impact-on-developer-productivity-and-happiness/

Questions?

Feedback (through anonymous form)

DESN2000 (SENG) 2026T2
Feedback Form

https://forms.office.com/r/kEcdPkQCaE

Feedback

- Email me directly:
  m.al-banna@unsw.edu.au

- Or talk to your mentor